LOGICAL REASONING

09/09/25

Series

Sequence of letters, numbers, words or objects

1. Number Series –

2, 5, 8, 11 → addition of 3

Examples –

E3. 32, 48, 120, ?, 450, 675

$32 \times 1.5 = 48$, $48 \times 2.5 = 120$

$120 \times 1.5 = 180$

Calendar

17/09/25

Doomsday Algorithm -

It allows us to calculate the day of a week using date.

* Developed in 1970 by John Conway.

What is doomsday?

Every year has doomsdays, days whose weekday is known.

Imp For every year the doomsday have same weekday.

Example - If Nov 7 (a doomsday) is Saturday, all other doomsday of the year would be Saturday as well.

* Each month has a fix doomsday

Table for doomsday -

Month	Date of doomsday	Memory hook
January	3rd (Normal year) 4th (Leap year)	3 in 3, 4 in 4
February	28th (Normal Year)	last day of Feb

	29th (leap year)	
March	7th March	7 Multiple (7, 14, 21, 28)
April	4th	4/4
May	9th	9 to 5
June	6th	6/6
July	11th	7-11
August	8th	8/8
September	5th	9 to 5
October	10th	10/10
November	7th	11-7
December	12th	12/12

Finding the Doomsday of a year –

Write the year as CCYY (century CC, year within century YY)

Steps –

	Step	What to do	Example for 2023
1.	Year	Take last two digits $Y = YY$	$2023 \rightarrow 23$
2.	Integer Division	Divide by 12 $a = Y // 12$	$23 // 12 = 1$ $a = 1$
3.	Remainder	Remainder (%) after division $b = Y \% 12$	$23 \% 12 = 11$ $b = 11$
4.	Integer division	Divide b by 4 $C = b // 4$	$11 // 4 = 2$ $C = 2$
5.	Sum	Add a, b, C $S = a + b + C$	$1 + 11 + 2 = 14$ $S = 14$
6.	Add Century Anchor	$a_n = 2$ for (2000 - 2099) $S_F = S + a_n$	$14 + 2 = 16$ $S_F = 16$
7.	Modulus by 7	Remainder by 7 $d_S = S_F \bmod 7$	$16 \bmod 7 = 2$ $d_S = 2$ (doomsday)

<u>Weekday Mapping (Standard Mapping)</u> -

0 = Sunday, 1 = Monday, 2 = Tuesday;

3 = Wednesday, 4 = Thursday, 5 = Friday
6 = Saturday

So, $\boxed{\text{Doomsday (2023)} = 2 = \text{Tuesday}}$

Century Anchor Values –

Centuries	Anchor	
1600 – 1699	2	Repeats (Start from leap year) (leap century)
1700 – 1799	0	
1800 – 1899	5	
1900 – 1999	3	
2000 – 2099	2	
2100 – 2199	0	
2200 – 2299 Repeats	5	
2300 – 2399	3	
2400 – 2499	2	
2500 – 2599	0	

* Pattern repeats after 400 years (leap century)

Formula –

$$\boxed{\text{Anchor} = (5 \times (C \bmod 4) + 2) \bmod 7}$$

C → Century Number (e.g 20 for 2000–209

For 2000 - 2099 - $C = 20$

$$\text{Anchor} = (5 \times (20 \bmod 4) + 2) \bmod 7$$
$$= (5 \times 0 + 2) \bmod 7$$
$$= 0 \bmod 7 + 2 \bmod 7$$
$$= 0 + 2 = 2$$

Using doomsday -

Example - 24 Dec 2023

- Doomsday (2023) Tuesday
- Anchor = 12 Dec = Tuesday
- 24 - 12 = 12 days
- 12 - 7 = 5 → 5 mod 7 = 5
- Tuesday + 5 = Sunday

Very Far Years -

Gregorian cycle repeats after 400 years

So -

- To find doomsday for 4000, reduce 4000 mod 400 = 0 → same as year 0 in the cycle = Tuesday
- For 7000 : 7000 mod 400 = 2
  Century 70

Same 7 steps - $(5 + (70 \% 4) + 2) \% 7$

$= (5 + 2 + 2) \% 7$

$= 9 \% 7 = 2 =$ Tuesday

Base year and later -

If we know doomsday for start of a century (or any base year) we can get for the next year, just by adding 1 each year after a leap year - always working mod 7.

Example - Starting at Year 2000.

Year	Rule	Doomsday
2000	Base (Tuesday = 2)	2
2001	+1	3 (Wed)
2002	+1	4 (Thu)
2003	+1	5 (Fri)
2004	+1 then again +1 (leap year)	7 mod 7 = 0 (Sunday)
2005	+1	1 (Monday)
⋮	+1 happens because 29 Feb adds an extra day.	
So on		

Reversal —

We need to subtract instead of adding.

For 1999 —

Starting point : Doomsday (2000) = 2 (Tuesday)

For 1999 — subtract 1 then again subtract 1 as 2000 is a leap year

ds = 2 − 1 − 1 = 0 (Sunday)

* This is true for all cases.

Rule of Thumb —

Going forward — +1 each year, +1 extra after a leap year

Going backward — −1 each year, −1 extra after a leap year

Number System

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Divisibility rules –

Divisibility by 2 –

If unit digit of number is 0, 2, 4, 6, 8 (an even number), then the number is divisible by 2.

Example – 42, 44 . . . 48, 334, 87640 → unit digit

Divisibility by 3 –

If the sum of digits of number is divisible by 3, then number is divisible by 3

Example – 75333 ⇒ 7+5+3+3+3 = 21 which is divisible by 3

Divisibility by 4 –

If last two digits of a number is 0 or divisible by 4 then the number is divisible by 4.

Example – 67824 → 24 is divisible by 4

6700, 54300, 27500 → Divisible by 4

Divisibility by 5 -

If the last digit of a number is either 0 or 5, then it is divisible by 5.

Example - 115, 67520 → divisible by 5

Divisibility by 6 -

If number is divisible by 2 and 3, then it is divisible by 6.

Example - 378, 444 → divisible by 2 & 3.

Divisibility by 7 -

If the difference between twice the last digit and the number formed by remaining digit is either 0 or a multiple of 7, then it is divisible by 7.

Example - 637 → 63 - (2x7) = 63 - 14 = 49

which is divisible by 7.

Divisibility by 8 -

If last 3 digit of number is divisible by 8 or 0, then the number is

divisible by 8.

Example – 2368 → 368 / 8 → 48 (divisible 368 by 8)

435000, 235000 → divisible by 8

Divisibility by 9 –

If sum of digits of a number is divisible by 9.

Example – 64737 → 6+4+7+3+7 → 27 (divisible by 9)

Divisibility by 11 –

Difference between sum of its digits at odd places and the sum of its digits at even places is 0 or multiple of 11.

Example – 9693926

$\Rightarrow (9+9+9+6) - (6+3+2)$

$\Rightarrow 33 - 11 = 22$, which is divisible by 11.

For Multiples of prime numbers –

If a number is divisible by prime factors of a number, then it is

divisible by that number too.

Example – $6 = 3 \times 2$ (If $n \rightarrow$ number to be divided, n is divisible by 3 and 2, then it is divisible by 6 too)

$12 = 3 \times 4$ (Same logic)
$14 = 7 \times 2$ $15 = 3 \times 5$

Note – This is valid for numbers more than 1 unique prime factor.

Divisibility by 13 –

Multiply unit place digit by 4, then add the product to the number formed by rest of digits. If result is 0 or multiple of 13, then it is divisible by 13.

Example – $182 \Rightarrow 18 + (2 \times 4) = 18 + 8 = 26$

$\therefore$ 182 divisible by 13

Divisibility by 16 –

Last 4 digits divisible by 4

Example – 8675224 → 5224 is divisible by 4

Divisibility by 17 –

Multiply unit place digit by 5, then subtract the product from the numbers left.

Example – 663 ⇒ 66 – (3 × 5) = 66 – 15 = 51

51 divisibly by 17

Divisibility by 25 & 125 –

25 → last 2 digits are 0 or divisible by 5

Example – 10075 → 75 divisible by 5

125 → last 3 digits are 0 or divisible by 125

Example – 13250 → 250 divisible by 125

Cyclicity and Unit digit -

Depends upon their units (ones) digits. Each unit digit shows a repetitive pattern which depends upon their powers.

Example — Cyclicity of 7

$7^1 = 7$ | unit digit = 7
$7^2 = 49$ | unit digit = 9
$7^3 = 343$ | unit digit = 3
$7^4 = 2401$ | unit digit = 1

Repeats after 4

$7^5 = 16807$ | unit digit = 7
$7^6 = 1176809$ | unit digit = 9
$7^7 = 823543$ | unit digit = 3
$7^8 = 5764801$ | unit digit = 1

Cyclic Nature

Cyclicity of digits -

Number	Cyclicity	Power Cycle
0	1	0
1	1	1
2	4	2, 4, 8, 6
3	4	3, 9, 7, 1
4	2	4, 6
5	1	5
6	1	6
7	4	7, 9, 3, 1
8	4	8, 4, 2, 6

Number	Cyclicity	Power Cycle
9	2	9, 1

Method to find unit digit –

1. If unit digit is 0, 1, 5, 6, then unit digit remains same

Example – $(12565)^{345}$ → Unit digit → 5

2. If unit digit is 2, 3, 4, 7, 8, 9

Step 1 :– Divide the power by 4 and find remainder –

Example – $(5767)^{35327}$

i) Divide – $35327 / 4 = 35324 \times 4 + 3$

Now, for 35324 i.e 4 cycle unit digit is 1 (for 7)

For the rest 3, we will use step 2

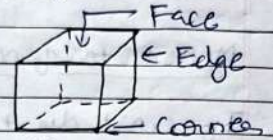
Step 2 :– Apply cyclicity rule on remainder –

Now, cyclicity of 7 is – 7, 9, 3, 1 (1 2 3)
so, 1 (previous ans) + 3 (i.e 3 places) is 3

Cubes and Dices

Cube – Shape with 6 square sides, all with the same size

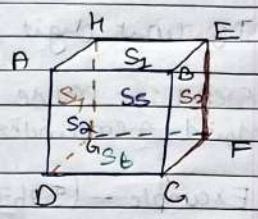
3-D shape having –



- 6 surface / faces
- 8 corners, 12 edges Fig – Cube

In a Cube – Length = Breadth = Height
$L = B = H$

Visible Faces –



ABEH – S_1, ABCD = S_2, BCEF – S_3

Invisible faces –

Fig – Cube with faces

ADGH = S_4, EFGH = S_5, CDGF = S_6

Cutting of a Cube or Cuboid –

After cutting a cube, following types of smaller cubes are obtained –

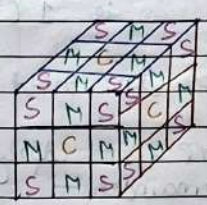


Fig: Type of Cut cubes

1. Corner Cubes = S (exist at each corner)
2. Middle Cubes = M (exist at the middle of each edge)

3. Central Cubes = C (exist at middle of each face)
4. Nucleus Cube / Inner Central Cube = N (hidden and exist at the centre of the larger cube)

Number of Cubes –

If a larger cube is cut into smaller cubes of equal volume so that each edge is divided into n parts (n−1) cuts –

Type of smaller cubes	Description	Formula	Explanation
1. Total Cubes	Total number of smaller cubes	n^3	n = edge of larger cube / edge of smaller cube
2. Inner / Nucleus cubes (N)	Cubes inside, not touching any outer face	$(n-2)^3$	This are completely inside, away from any face
Paint → 0 (No paint Faces)			
3. Central Cube (C)	Cubes on the faces, but not on edges or corners	$6(n-2)^2$	Found on each face, excluding edges and
Paint → 1 (one paint face)			

	Type	Description	Formula	Explanation
4	Middle Cubes (M) (2 painted Faces)	Cubes on the edges, not at corners	$12(n-2)$	Found in a line along the edges, excluding corners
5	Corner Cubes (S) (3 painted faces)	Cubes at the corners of Cube	8	Always 8 corners in a cube (This is fixed)

Paint → (2 painted Faces)

Paint → (3 painted faces)

For Cube painted and cut –

A larger cube is painted with a single colour, and then it is cut into smaller cubes. Using the paint formula we can find number of such cubes

Cutting of Cuboid –

When cuboid is cut into smaller cubes of equal volume, then

Total no. of cubes = $\frac{\text{Volume of cuboid}}{\text{Volume of smaller cube}}$

Volume of Cuboid = length × breadth × height (Cuboid)
volume of smaller cube = (side of smaller cube)^3

Dice – 3D figure with six faces, each displaying different numbers, letters or colours.

Types of Dice –

1. Standard Dice – No. of dots on opposite faces is addition of 7.

Number Arrangement –

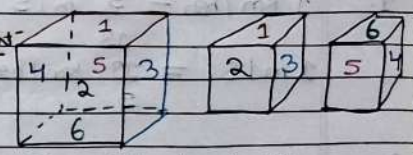


Fig:– Standard dice

- 1 is opposite to 6 {1+6 = 7}
- 2 is opposite to 5 {2+5 = 7}
- 3 is opposite to 4 {3+4 = 7}

In a dice, two opposite faces will never be seen together. Like 1 and 6 can never appear together.

- This can also be used to find if a dice is standard dice or not.
- Opposite faces are very useful in solving dice problems.

Time, speed and Distance

26/09/25

Formula –

1. Distance = Speed × Time

2. Time = $\frac{\text{Distance}}{\text{Speed}}$

3. Speed = $\frac{\text{Distance}}{\text{Time}}$

Conversion of Units –

1. 1 km/hr = $\frac{5}{18}$ m/s

   1 m/s = $\frac{18}{5}$ km/hr

2. 1 km = 1000 m, 1 hr = 60 min, 1 min = 60 sec.

3. 1 mile = 1609.3 m = 1.6093 km

4. 1 km = 0.621 mile, 1 yard = 0.9144 m,
   1 m = 1.0936 yards

5. 1 m = 39.4 inches, 1 inch = 0.0253 m